AM-Align: Globally Optimal Estimation of Accelerometer–Magnetometer Misalignment

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***Abstract*— Misalignment estimation of the accelerometer–magnetometer (AM) combination is crucial for accurate attitude and position estimation in various applications. Despite significant research efforts over the past decades, existing algorithms rely on sufficient and outlier-free data to guarantee a satisfactory initial guess for optimization. This article presents AM-Align, a novel globally optimal method that addresses this limitation. The key innovation of AM-Align is its ability to find all global optima by identifying the solution set with the least loss function values using the polynomial eigenvalue method, making it independent of initial guesses and robust to insufficient data and outliers. Simulation studies demonstrate the complete solution set for the minimal case of the AM calibration problem, providing valuable insights. Experimental results showcase the superiority of AM-Align compared to state-of-the-art methods, requiring fewer measurements while achieving higher accuracy and robustness. Notably, AM-Align outperforms representative algorithms by a significant margin in terms of calibration precision and resilience to data scarcity and outliers. This work presents a significant advancement in AM calibration, enabling more reliable attitude and position estimation in real-world scenarios. We will make the AM-Align codebase (https://github.com/JokerJohn/AM_Align) publicly available to the community, further advancing the development of this field.**

***Index Terms*— Accelerometer, attitude estimation, magnetometer, misalignment calibration, sensor alignment.**

I. Introduction

A. Motivation and Challenges

INERTIAL sensors, including gyroscopes and accelerometers, are essential for state estimation and sensing in various applications such as robotics, wearable devices, and mechatronics. Magnetometers, as aiding sensors that measure the nearby magnetic field of a rigid body, significantly contribute to drift-free estimation and feedback-based actuation [1], [2], [3]. The fusion of inertial sensors and magnetometers has been widely studied, with the accelerometer–magnetometer (AM) combination forming a fundamental mechanism for attitude estimation [4], [5]. However, the calibration and alignment of these sensors, particularly low-cost microelectromechanical system (MEMS) sensors, remain challenging. MEMS accelerometers and magnetometers are prone to errors such as scale factors, nonorthogonality, and biases, which require careful calibration. Moreover, misalignments between the accelerometer and magnetometer can occur during the MEMS production process, leading to errors in attitude estimation when fusing raw measurements. This problem, known as the AM alignment (AMA), as shown in Fig. 1, has been the focus of extensive research in recent years.

Existing calibration methods for accelerometers and magnetometers often rely on the principle that true values are distributed on a standard sphere, leading to the development of ellipsoidal fitting techniques. While precision calibration using accurate turntables can align the sensor frames to a standard frame, such equipment is expensive and may not be readily accessible for consumer-level products. Therefore, there is a need for easy-to-implement calibration methods that can be performed by users without specialized equipment. Despite the efforts to develop calibration and alignment techniques, several challenges persist in this field.

To address these limitations, we propose the AM-Align method, which aims to provide a globally optimal solution for the AMA problem. AM-Align is independent of initial guesses and robust to insufficient and outlier-corrupted data, requiring only four pairs of measurements for complete calibration based on an approximate preliminary geodetic position, which is less than state-of-the-art approaches. By exploiting the algebraic structure of the sensor measurement models, AM-Align formulates the alignment problem as a constrained least-squares optimization in unit quaternion space and solves it efficiently using the polynomial eigenvalue technique.

B. Related Work

The AMA problem has been studied extensively in recent years, with various approaches proposed to address the challenges in calibration and sensor fusion. Early works focused on estimating the ego-misalignment between multiple inertial sensor arrays [6], [7]. Li et al. [8] provided an overview

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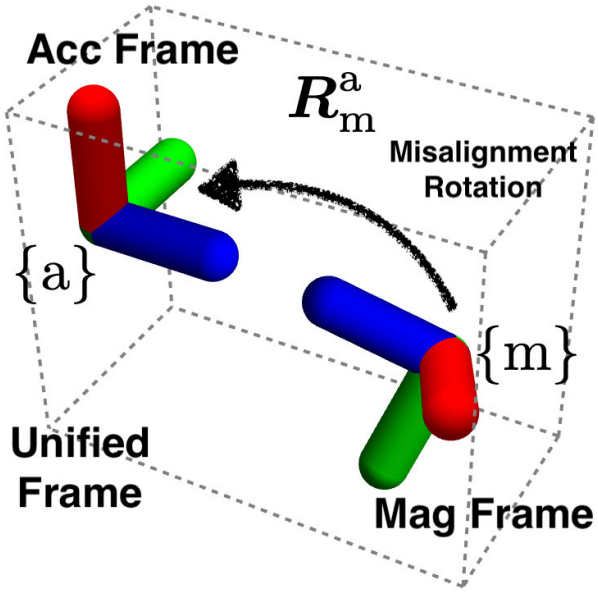


Fig. 1. Misalignment calibration matrix (rotation $\boldsymbol{R}_m^a$ between accelerometer $\{a\}$ and magnetometer $\{m\}$ frames). The red, green, and blue axes denote the frames in the Cartesian $Z-Y-X$ sequence.

of magnetic sensors for navigation applications, discussing the challenges and solutions for calibration and alignment. In the context of sensor fusion, Zhang et al. [9], [10] proposed quaternion-based Kalman filters with vector selection to improve the accuracy and robustness of orientation estimation. Papafotis and Sotiriadis [11], [12], [13], [14], [15] addressed the problem of orientation sensor fusion and calibration for motion tracking in mobile devices. They developed fusion algorithms that combine data from inertial sensors and magnetometers while considering misalignment and calibration parameters, demonstrating improved accuracy and robustness on real-world datasets. Regarding calibration techniques, Vasconcelos et al. [16] presented a geometric approach to magnetometer calibration in the sensor frame using ellipsoid fitting to estimate the bias, scale factors, and misalignment. Wu and Shi [17] developed a maximum likelihood estimator for calibrating three-axis magnetometers, while Wu and Luo [18] proposed efficient algorithms for aligning inertial and magnetic sensors. The inertial-magnetic alignment scheme has also been applied to autonomous aerial systems [19] and generalized sensors [20].

Despite these advancements, the field of accelerometer and magnetometer calibration and misalignment estimation still faces several challenges. Many algorithms require long-duration motions in different directions to achieve observable estimates, often necessitating specific sensor motion trajectories designed by practitioners to satisfy the calibration procedure. Furthermore, some methods heavily rely on analytical initial guesses, which may be noisy and inaccurate, potentially leading to divergent results after optimization refinement. Certain motion patterns are also required to guarantee the convergence of the algorithmic computation.

C. Contributions

The main contributions of this article are summarized as follows.

1) We propose AM-Align, a globally optimal method for solving the AMA problem. AM-Align is independent of initial guesses and robust to insufficient and outlier-corrupted data, requiring only four pairs of measurements for complete calibration, which is less than state-of-the-art approaches.
2) By exploiting the algebraic structure of sensor measurement models, we formulate the alignment problem as a constrained least-squares optimization in unit quaternion space. This problem is further reduced to a system of multivariate polynomial equations and solved efficiently using the polynomial eigenvalue technique.
3) Extensive simulations and real-world experiments demonstrate that AM-Align consistently achieves higher accuracy and robustness compared to state-of-the-art approaches under different conditions.

D. Organization

The remainder of this article is structured as follows. Section II presents the problem formulation and the proposed globally optimal solution. Section III provides experimental validation and comparisons with state-of-the-art methods. Finally, Section IV concludes this article.

II. Problems and Solution

In this section, we present the problem formulation and the proposed globally optimal solution for AM calibration. We begin by introducing the sensor model and the preliminaries of misalignment calibration (see Sections II-A and II-B). Then, we propose a solution strategy for this AMA problem in Section II-C.

A. Preliminary: Sensor Model

For a vector-field sensor, the raw readings can be modeled as follows:

$$\boldsymbol{v}^b = \boldsymbol{S}\boldsymbol{R}\boldsymbol{v}^r + \boldsymbol{b}_v + \boldsymbol{\epsilon}_v \tag{1}$$

where $\boldsymbol{v}^b$ and $\boldsymbol{v}^r$ are 3-D vector measurements in the body frame and reference frame, respectively. The calibration matrix $\boldsymbol{S} \in \mathbb{R}^{3\times3}$ accounts for scale factor and nonorthogonality. The rotation matrix $\boldsymbol{R}$ belongs to the special orthogonal group $\mathrm{SO}(3) := \{\boldsymbol{R} \in \mathbb{R}^{3\times3} | \boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{I}, \det(\boldsymbol{R}) = 1\}$. The terms $\boldsymbol{b}_v$ and $\boldsymbol{\epsilon}_v \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_v)$ represent the constant bias and stochastic noise, respectively. The noise covariance matrix $\boldsymbol{\Sigma}_v$ is typically modeled as isotropic, $\boldsymbol{\Sigma}_v = \gamma^2\boldsymbol{I}$, where γ denotes the variance or noise level.

In this article, r-frame is the north–east–down (NED) reference frame. The *intrinsic* calibration problem of the vector-field sensor is to estimate unknown parameters $\boldsymbol{S}, \boldsymbol{R}, \boldsymbol{b}_v$ and sometimes, $\boldsymbol{v}^r$, with given measurements of $\boldsymbol{v}^b$. For the accelerometer, $\boldsymbol{v}^r$ is known and can be set to $\boldsymbol{v}^r = (0, 0, 9.8)^\top$ m/s^2 in the NED frame. For magnetometer, however, since $\boldsymbol{v}^r$ is the function of local geodetic coordinates, $\boldsymbol{v}^r$ should be estimated as well. Intrinsic calibration of magnetometers has been extensively studied. Thus, the practitioners can employ methods,

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e.g., [17], [21], [22] for *ad hoc* calibration prior to specific use.

B. Proposed Misalignment Calibration

After a proper intrinsic calibration procedure, the accelerometer and magnetometer in the nondistorted condition have the following measurements related by the rotation matrix $\boldsymbol{R} \in \mathrm{SO}(3)$:

$$\boldsymbol{a}^b = \boldsymbol{R}\boldsymbol{a}^r, \quad \boldsymbol{m}^b = \boldsymbol{R}\boldsymbol{m}^r \tag{2}$$

in which $\boldsymbol{a}^b$ and $\boldsymbol{m}^b$ are **normalized** measurements in the body frame from the accelerometer and magnetometer, respectively; $\boldsymbol{a}^r = (0, 0, 1)^\top$ and $\boldsymbol{m}^r = (m_N, 0, m_D)^\top$ denote the corresponding normalized vectors in the NED frame, while m_N and m_D are components in the north and down directions, respectively. A fact in these sensor models is that

$$\boldsymbol{a}^b \cdot \boldsymbol{m}^b = \left(\boldsymbol{a}^r\right)^\top \boldsymbol{R}^\top \boldsymbol{R}\boldsymbol{m}^r = \left(\boldsymbol{a}^r\right)^\top \boldsymbol{m}^r = \boldsymbol{a}^r \cdot \boldsymbol{m}^r \tag{3}$$

which is the dot-product equality for attitude determination from two vector observations, which has been extensively adopted in calibration and attitude determination [23], [24], [25]. Expanding (3) gives

$$\boldsymbol{a}^b \cdot \boldsymbol{m}^b = \left\|\boldsymbol{a}^r\right\| \left\|\boldsymbol{m}^r\right\| \cos\theta = \cos\theta \tag{4}$$

where θ is the local magnetic dip angle. For common AM combinations, the alignment between the two sensors may not be refined. Therefore, when using the two sensors for attitude estimation, the frame misalignment between them must be determined. Denoting the rotation from the magnetometer to the accelerometer as $\boldsymbol{R}_m^a$, it is able to relate the measured normalized magnetic field $\tilde{\boldsymbol{m}}^b$ in the magnetometer frame and corresponding $\boldsymbol{m}^b$ in the accelerometer frame by $\boldsymbol{m}^b = \boldsymbol{R}_m^a \tilde{\boldsymbol{m}}^b$. Therefore, (4) becomes

$$\boldsymbol{a}^b \cdot \boldsymbol{m}^b = \left(\boldsymbol{a}^b\right)^\top \boldsymbol{R}_m^a \tilde{\boldsymbol{m}}^b = \cos\theta. \tag{5}$$

From (5), one can observe that there are two unknowns, i.e., $\boldsymbol{R}_m^a$ and θ. Thus, the task of the AMA is to estimate $\boldsymbol{R}_m^a$ and θ from multiple-position measurements of $\boldsymbol{a}^b$ and $\boldsymbol{m}^b$ in a simultaneous manner. The kernel optimization is expressed as follows:

$$\underset{\boldsymbol{R}_m^a \in \mathrm{SO}(3), \theta \in \mathbb{R}}{\arg\min} \quad \mathcal{L} = \sum_{i \in \mathcal{I}} \left[\left(\boldsymbol{a}_i^b\right)^\top \boldsymbol{R}_m^a \tilde{\boldsymbol{m}}_i^b - \cos\theta \right]^2 \tag{6}$$

in which i denotes the sample index of accelerometer and magnetometer measurements, while $\mathcal{I}$ indicates all indices of available inliers. Optimization (6) can be interpreted as

$$\underset{\boldsymbol{R}_m^a \in \mathrm{SO}(3), c \in \mathbb{R}}{\arg\min} \quad \mathcal{L} = \sum_{i \in \mathcal{I}} \left[\left(\boldsymbol{a}_i^b\right)^\top \boldsymbol{R}_m^a \tilde{\boldsymbol{m}}_i^b - c \right]^2, \quad \text{s.t. } |c| \le 1. \tag{7}$$

In [14], an analytical algorithm is designed to compute closed-form approximation of $\boldsymbol{R}_m^a$ and c from (7). Then, the gradient-descent algorithm is applied to solve a refined solution from the closed-form initial guess. The method in [14] is practical and accurate but can only be feasible in the condition that the following matrix has full rank:

$$\boldsymbol{A} = \begin{bmatrix} \left(\tilde{\boldsymbol{m}}_1^b \otimes \boldsymbol{a}_1^b\right)^\top \\ \left(\tilde{\boldsymbol{m}}_2^b \otimes \boldsymbol{a}_2^b\right)^\top \\ \cdots \\ \left(\tilde{\boldsymbol{m}}_i^b \otimes \boldsymbol{a}_i^b\right)^\top \\ \cdots \end{bmatrix}, \quad i \in \mathcal{I}. \tag{8}$$

If $\boldsymbol{A}$ is rank-deficient, the estimated approximation of $\mathrm{vec}(\boldsymbol{R}_m^a)$ will not be a proper initial guess and may diverge to a local minimum in a further gradient-descent search. To guarantee the rank of $\boldsymbol{A}$, at least nine different pairs of measurements are required. However, from (7), one can conclude that $\boldsymbol{R}_m^a$ and c are observable in the condition that there are at least four pairs of measurements. This is because the rotation matrix $\boldsymbol{R}_m^a$ has three degrees of freedom. This article solves the problem (7) without a closed-form initial guess, but, rather by using the Gröbner-basis method, the global optimum can be sought with high accuracy. Choosing $\boldsymbol{q}_m^a$ as the quaternion of $\boldsymbol{R}_m^a$, it is able for us to write the gradient of $\mathcal{L}$

$$\nabla_{\boldsymbol{y}} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{y}} \tag{9}$$

where $\boldsymbol{y} = [(\boldsymbol{q}_m^a)^\top, c]^\top$. All local minima and maxima occur in the condition of

$$\nabla_{\boldsymbol{y}} \mathcal{L} = \boldsymbol{0}. \tag{10}$$

By extending $\nabla_{\boldsymbol{y}} \mathcal{L} = \boldsymbol{0}$, one obtains

$$c = \frac{1}{\mathcal{N}} \sum_{i \in \mathcal{I}} \left(\boldsymbol{a}_i^b\right)^\top \boldsymbol{R}_m^a \tilde{\boldsymbol{m}}_i^b. \tag{11}$$

Inserting (11) into the original objective $\mathcal{L}$ gives a new optimization objective $\hat{L}$ without c, say

$$\underset{\|\boldsymbol{q}_m^a\|=1}{\arg\min} \ \hat{L}\left(\boldsymbol{q}_m^a\right). \tag{12}$$

The optimization can be solved via constructing the Lagrangian such that

$$\tilde{L} = \hat{L} + \lambda \left[\left(\boldsymbol{q}_m^a\right)^\top \boldsymbol{q}_m^a - 1 \right] \tag{13}$$

where λ is the Lagrange multiplier. The quaternion and Lagrange multiplier can be sought by means of

$$\nabla_{\boldsymbol{\zeta}} \tilde{L} = \boldsymbol{0} \tag{14}$$

where $\boldsymbol{\zeta} = [(\boldsymbol{q}_m^a)^\top, \lambda]^\top$. This will construct a polynomial system of five subequations. Each equation has different sets of monomials. The first four subequations are essential for the solution of $\boldsymbol{q}_m^a$. The fifth one is simply the quaternion unit norm condition. If we parameterize quaternion as $\boldsymbol{q}_m^a = (q_0, q_1, q_2, q_3)^\top$, the first four sets of monomials can be summarized as

$$\boldsymbol{m}_1 = \begin{pmatrix} q_0^3, q_0^2 q_1, q_0^2 q_2, q_0^2 q_3, q_0 q_1^2, q_0 q_1 q_2, q_0 q_1 q_3, \\ q_0 q_2^2, q_0 q_2 q_3, q_0 q_3^2, \lambda q_0, q_1^3, q_1^2 q_2, q_1^2 q_3, q_1 q_2^2, \\ q_1 q_2 q_3, q_1 q_3^2, q_2^3, q_2^2 q_3, q_2 q_3^2, q_3^3 \end{pmatrix}^\top \tag{15a}$$

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$$\boldsymbol{m}_2 = \begin{pmatrix} q_0^3, q_0^2q_1, q_0^2q_2, q_0^2q_3, q_0q_1^2, q_0q_1q_2, q_0q_1q_3, \\ q_0q_2^2, q_0q_2q_3, q_0q_3^2, q_1^3, q_1^2q_2, q_1^2q_3, q_1q_2^2, \\ q_1q_2q_3, q_1q_3^2, \lambda q_1, q_2^3, q_2^2q_3, q_2q_3^2, q_3^3 \end{pmatrix}^\top \tag{15b}$$

$$\boldsymbol{m}_3 = \begin{pmatrix} q_0^3, q_0^2q_1, q_0^2q_2, q_0^2q_3, q_0q_1^2, q_0q_1q_2, q_0q_1q_3, \\ q_0q_2^2, q_0q_2q_3, q_0q_3^2, q_1^3, q_1^2q_2, q_1^2q_3, q_1q_2^2, \\ q_1q_2q_3, q_1q_3^2, q_2^3, q_2^2q_3, q_2q_3^2, \lambda q_2, q_3^3 \end{pmatrix}^\top \tag{15c}$$

$$\boldsymbol{m}_4 = \begin{pmatrix} q_0^3, q_0^2q_1, q_0^2q_2, q_0^2q_3, q_0q_1^2, q_0q_1q_2, q_0q_1q_3, \\ q_0q_2^2, q_0q_2q_3, q_0q_3^2, q_1^3, q_1^2q_2, q_1^2q_3, q_1q_2^2, \quad c \\ q_1q_2q_3, q_1q_3^2, q_2^3, q_2^2q_3, q_2q_3^2, q_3^3, \lambda q_3 \end{pmatrix}^\top. \tag{15d}$$

The coefficients rely on input measurements, whose analytical forms can be given by symbolic engines of mathematical software such as Python `SymPy` or Mathworks `MATLAB`. From these monomials, we can see that they have common parts but are not the same for the rest parts. All these monomials form a cubic polynomial system to be solved. To obtain all local minima of $\boldsymbol{q}_m^a$, one needs to solve all solutions of $\boldsymbol{q}_m^a$ and λ. This indicates that we are dealing with a five-variable cubic polynomial system. There is no closed-form solution to such a high-order polynomial system according to the Abel–Ruffini theorem. However, by using Bezout's theorem [26], we can obtain the maximum solution count of this system, i.e., it has at most $5 \times 4^{(3-1)} = 80$ distinct solutions. Although, according to Bezout's theorem, we have concluded that our system to be solved has at most 80 distinct solutions, since one quaternion and its negative represent the same rotation, there are at most 40 solutions to be found. Note that not all these 40 solutions are in the real space. Therefore, only real solutions will be considered, which normally covers 4∼16 real solutions in engineering practice. After all real local quaternion critical points are obtained, inserting them back into (11) gives multiple $s = \cos\theta$. Then, inserting these quaternions along with s critical points back into the original objective function $\mathcal{L}$ produces a number array, with which we obtain the globally optimal solution corresponding to the least objective function value after a typical sort algorithm. In the following, we specifically present how we obtain all 40 solutions using the polynomial eigenvalue solver.

C. Solution Strategies

The polynomial eigenvalue method is a rigorous mathematical approach for solving systems of polynomial equations of the form $f(\boldsymbol{x}) = \boldsymbol{0}$, where f is a polynomial function and $\boldsymbol{x}$ is an unknown vector. This method leverages the connection between polynomial systems and eigenvalue problems to find the solutions by reformulating the polynomial system as a generalized eigenvalue problem. To formally describe the polynomial eigenvalue method, let $\mathcal{K}[x_1, x_2, \ldots, x_n]$ be the polynomial ring over a field $\mathcal{K}$, where $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)^\top$ constitute the unknown vector. Consider the polynomial system $f(\boldsymbol{x}) = [f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})]^\top = \boldsymbol{0}$, where each $f_i(\boldsymbol{x})$ is a polynomial in $\mathcal{K}[x_1, x_2, \ldots, x_n]$.

The first step in the polynomial eigenvalue method is to linearize the polynomial system. This is achieved by introducing new variables $z_1, z_2, \ldots, z_k$ to represent the monomials in the polynomial system. Let $\boldsymbol{\varphi}$ be the vector of all monomials in the polynomial system, and define the linearization map $\pi : \mathcal{K}[x_1, x_2, \ldots, x_m] \to \mathcal{K}^k$ as $\pi(x_i) = \boldsymbol{e}_i$, where $\boldsymbol{e}_i$ is the ith standard basis vector in $\mathcal{K}^k$. The linearized system can be written as $\boldsymbol{H}\boldsymbol{z} = \boldsymbol{0}$, where $\boldsymbol{H}$ is an $m \times k$ matrix called the companion matrix, and $\boldsymbol{z} = (z_1, z_2, \ldots, z_k)^\top$ is the vector of new variables.

Next, we construct a matrix $\boldsymbol{G} \in \mathcal{K}^{k\times k}$ that encodes the relationships between the original variables $\boldsymbol{x}$ and the new variables $\boldsymbol{z}$. The matrix $\boldsymbol{G}$ is typically a block diagonal matrix, with each block corresponding to a particular variable in the original system. The entries of $\boldsymbol{G}$ are determined by the linearization map π and the structure of the polynomial system. The generalized eigenvalue problem associated with the polynomial system $f(\boldsymbol{x}) = \boldsymbol{0}$ is given by $\boldsymbol{H}\boldsymbol{\varrho} = \mu\boldsymbol{G}\boldsymbol{\varrho}$, where $\mu \in \mathcal{K}$ is an eigenvalue and $\boldsymbol{\varrho} \in \mathcal{K}^k$ is a corresponding eigenvector. The solutions to the polynomial system can be obtained from the eigenvectors $\boldsymbol{\varrho}$ by considering the appropriate components that correspond to the original variables $\boldsymbol{x}$.

To solve the generalized eigenvalue problem $\boldsymbol{H}\boldsymbol{\varrho} = \mu\boldsymbol{G}\boldsymbol{\varrho}$, we employ numerical linear algebra techniques such as the QZ algorithm or the Jacobi–Davidson method. These methods compute the eigenvalues μ and corresponding eigenvectors v, providing solutions to the original polynomial system. The polynomial eigenvalue method has several advantages, including the ability to handle systems with a large number of variables and high-degree polynomials, exploit the sparsity of the polynomial system, and leverage well-established numerical linear algebra techniques. However, it also has some limitations, such as the increase in problem size due to the linearization step, the potential for large and dense companion matrices, and sensitivity to numerical instabilities. This method has been successfully applied to various problems in engineering, physics, and other fields. It has been implemented in several software packages, such as MATLAB's Polynomial Toolbox and the Julia package `PolynomialRoots.jl`. Thus, we use MATLAB `mupad` symbolic engine to derive the symbolic manipulation of constructing matrices $\boldsymbol{H}$ and $\boldsymbol{G}$. Specifically, we use `equationsToMatrix` function to extract coefficients for $\boldsymbol{H}$ and $\boldsymbol{G}$. The entire algorithmic procedure is summarized in Algorithm 1.

III. Experimental Results

In this section, we first investigate the performance of AM-Align in the minimal case scenario (see Section III-A1) and conduct Monte Carlo simulations (see Section III-A2) to assess its robustness to noise and measurement errors. Then, we validate the effectiveness of AM-Align in real-world applications using handheld devices (see Section III-B2) and microaerial vehicles (see Section III-B3).

A. Simulation Studies

1) Minimal Case Analysis: Typically, for previous candidates, at least nine pairs of measurements are required for a complete misalignment solution, which can be seen in [11, Fig. 3], [14, Fig. 2], and [19, Fig. 2]. When we have

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Algorithm 1 Accelerometer–Magnetometer Alignment

Require: Accelerometer measurements $\boldsymbol{a}_i^b \ {}_{i=1}^{N}$, magnetometer measurements $\tilde{\boldsymbol{m}}_i^b \ {}_{i=1}^{N}$

Output: Rotation matrix $\boldsymbol{R}_{\mathrm{m}}^{\mathrm{a}}$, magnetic dip angle θ
Construct the least-squares problem according to (7)
Solve for c analytically using (11)
Formulate the polynomial system $f(\boldsymbol{x}) = \boldsymbol{0}$ based on (14) and (15)
Linearize the polynomial system by introducing variables $z_j \ {}_{j=1}^{k}$
Construct the companion matrix $\boldsymbol{H} \in \mathbb{R}^{m\times k}$ and the coefficient matrix $\boldsymbol{G} \in \mathbb{R}^{k\times k}$
Solve the generalized eigenvalue problem $\boldsymbol{H}\boldsymbol{\varrho} = \mu\boldsymbol{G}\boldsymbol{\varrho}$
Extract eigenvectors $\boldsymbol{\varrho}_j \ {}_{j=1}^{40}$ and retrieve quaternions $\boldsymbol{q}_{\mathrm{m,j}}^{\mathrm{a}} \ {}_{j=1}^{40}$
for $j = 1$ to 40 **do**
    Calculate the magnetic dip angle candidate s_j using (11)
    Evaluate the objective function $\mathcal{L}(\boldsymbol{q}_{\mathrm{m,j}}^{\mathrm{a}}, s_j)$ and add to the array $\{\mathcal{L}\}$
end for
Sort $\{\mathcal{L}\}$ and find the least one $\mathcal{L}_{\min}$.
$(\boldsymbol{q}_{\mathrm{m,opt}}^{\mathrm{a}}, s_{\mathrm{opt}}) \leftarrow argmin_j\, \mathcal{L}_{\min}(\boldsymbol{q}_{\mathrm{m,j}}^{\mathrm{a}}, s_j)$
$\boldsymbol{R}_{\mathrm{m}}^{\mathrm{a}} \leftarrow \mathrm{quaternionToRotationMatrix}(\boldsymbol{q}_{\mathrm{m,opt}}^{\mathrm{a}})$
$\theta \leftarrow \arccos(s_{\mathrm{opt}})$
return $\boldsymbol{R}_{\mathrm{m}}^{\mathrm{a}}$, θ

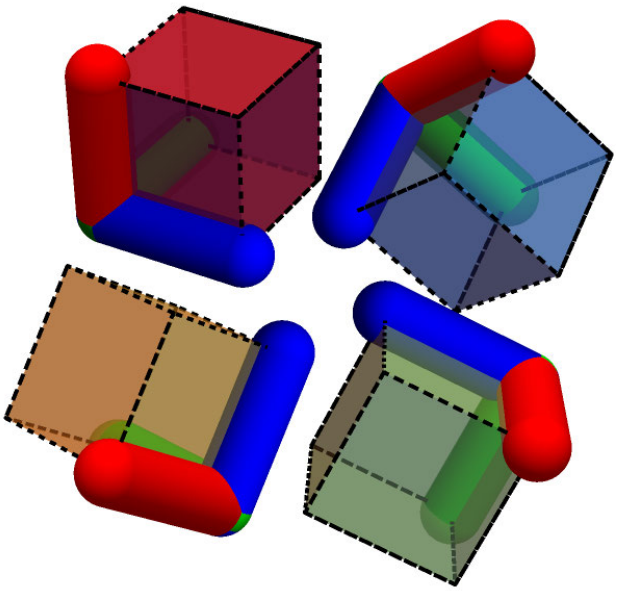


Fig. 2. Minimal sensor placements required for the proposed method. The red, green, and blue axes denote the Euler angles in roll, pitch, and yaw, respectively. Only four different poses are needed for a complete calibration, which is less than that in existing representatives.

$\mathcal{N} \geq 4$ pairs of AM measurements, the polynomial system in (14) becomes solvable. Fig. 2 shows an example of sensor placements, which requires only four different poses to complete the entire calibration process. Each pose denotes a different sensor placement in its corresponding direction,

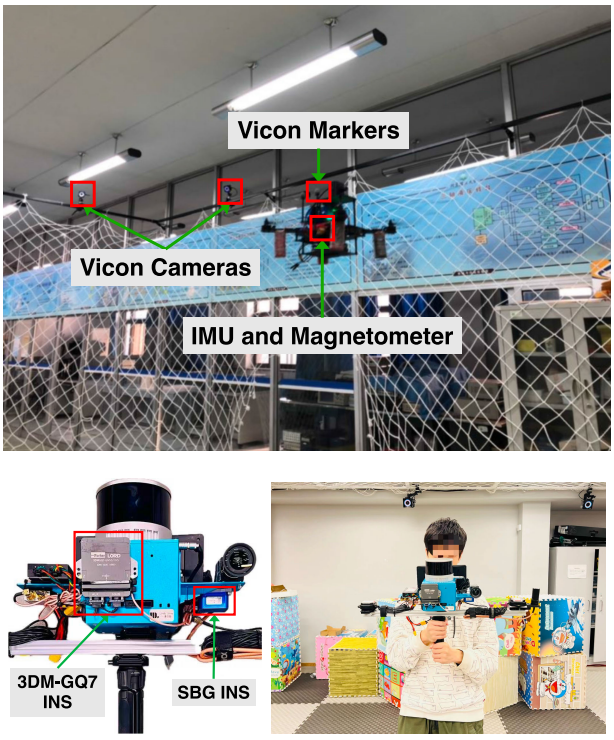


Fig. 3. Autonomous aerial vehicle platform used for hardware synthesis and verification (top). The platform is tested in a laboratory with the Vicon motion-capture ground-truth system. Handheld sensor suite featuring two different combinations of magnetometers and accelerometers (bottom left). Calibration experiments conducted by the researcher in a magnetically undisturbed room, utilizing the handheld suite across various motion directions (bottom right).

representing a certain pair of nonrepetitive measurements of the accelerometer and the magnetometer. Note that these poses are not rigorously required to be continuous. However, there shall be some differences between various poses in order to guarantee the solvability of the optimization problem. We present our experimental sensor suite for validation in Fig. 3, whose details will be given in later subsections. A typical pose sequence of the calibration process is depicted in Fig. 4 where four pairs of distinct poses are presented.

When exactly $\mathcal{N} = 4$, which is the minimal case of the studied problem, it can be empirically obtained that there are at most six ambiguous optimal solutions of a total of 40 distinct ones simultaneously satisfying the polynomial system. Sometimes, the minimal system has four ambiguous optimal solutions, which depends on the parameter configuration. We call this phenomenon the mirror effect of the solutions. If $\mathcal{N} = 5$, the fifth pair of measurements helps with verifying which of the ambiguous solutions is correct. To be clear, we present here a numerical example showing how the system is solved in the minimal case. We simulate the true misalignment rotation as

$$\boldsymbol{R}_{m,\mathrm{true}}^{a} = \begin{pmatrix} 0.8698 & 0.2063 & 0.4483 \\ -0.2507 & 0.9672 & 0.04127 \\ -0.4251 & -0.1483 & 0.8929 \end{pmatrix}$$

with $\cos\theta_{\mathrm{true}} = 0.861311153421915$, which denotes a position in the Northern hemisphere. The normalized accelerometer and magnetometer measurement pairs are simulated using

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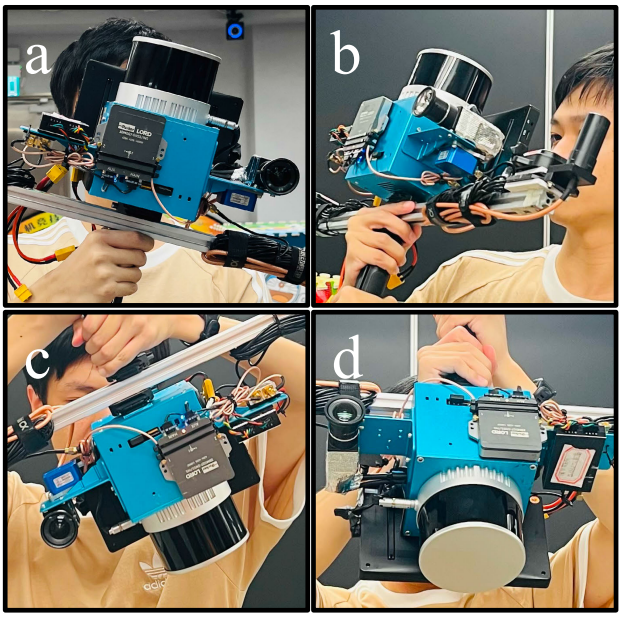


Fig. 4. Experimental validation of our proposed algorithm across four distinct poses (a)–(d) real-world scenarios, demonstrating its comprehensive effectiveness.

model (2) with the noise level of 10^{-3} for the accelerometer with the unit of m/sec^2 and for magnetometer with a unit of mGauss as follows:

$$\begin{aligned}
\boldsymbol{a}_1^b &= (-0.32646898, 0.93167874, 0.15935094)^\top \\
\boldsymbol{a}_2^b &= (0.94024823, -0.24908176, 0.23214553)^\top \\
\boldsymbol{a}_3^b &= (-0.078698444, -0.040278934, -0.99608442)^\top \\
\boldsymbol{a}_4^b &= (-0.30214477, -0.55269862, -0.77668061)^\top \\
\boldsymbol{m}_1^b &= (-0.26457266, 0.88781747, -0.37653877)^\top \\
\boldsymbol{m}_2^b &= (0.56789277, 0.39065359, 0.72449125)^\top \\
\boldsymbol{m}_3^b &= (0.15855473, -0.39080406, -0.90671527)^\top \\
\boldsymbol{m}_4^b &= (0.12203279, 0.018472239, -0.99235416)^\top
\end{aligned}$$

with which we construct the optimization kernel and solve it using the proposed method. The proposed algorithm returns six pairs of distinct solutions

$$\boldsymbol{R}_{m,1}^a = \begin{pmatrix} -0.4789 & 0.1937 & 0.8562 \\ 0.8354 & 0.4001 & 0.3768 \\ -0.2696 & 0.8958 & -0.3534 \end{pmatrix}, \quad \cos\theta_1 = 0.1598$$

$$\boldsymbol{R}_{m,2}^a = \begin{pmatrix} 0.8685 & 0.2078 & 0.4499 \\ -0.2523 & 0.9668 & 0.0405 \\ -0.4266 & -0.1486 & 0.8921 \end{pmatrix}, \quad \cos\theta_2 = 0.8612$$

$$\boldsymbol{R}_{m,3}^a = \begin{pmatrix} -0.9555 & -0.0186 & 0.2944 \\ -0.2707 & -0.3416 & -0.9000 \\ 0.1173 & -0.9397 & 0.3214 \end{pmatrix}, \quad \cos\theta_3 = -0.0982$$

$$\boldsymbol{R}_{m,4}^a = \begin{pmatrix} -0.0534 & 0.4665 & -0.8829 \\ 0.9372 & -0.2818 & -0.2056 \\ -0.3447 & -0.8384 & -0.4222 \end{pmatrix}, \quad \cos\theta_4 = -0.7191$$

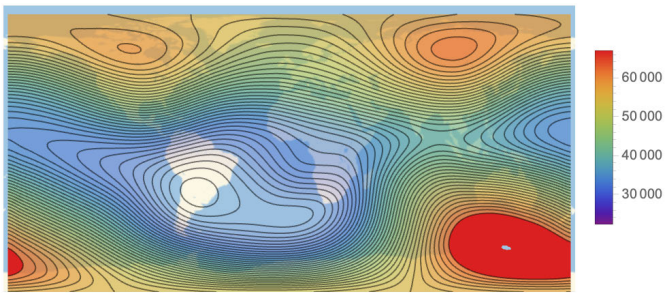


Fig. 5. Geomagnetic total intensity over the world map in the altitude of the sea level.

$$\boldsymbol{R}_{m,5}^a = \begin{pmatrix} 0.5082 & -0.5002 & -0.7011 \\ -0.4606 & -0.8457 & 0.2694 \\ -0.7277 & 0.1860 & -0.6602 \end{pmatrix}, \quad \cos\theta_5 = -0.4812$$

$$\boldsymbol{R}_{m,6}^a = \begin{pmatrix} -0.8918 & -0.3134 & 0.3264 \\ -0.1718 & -0.4328 & -0.8850 \\ 0.4186 & -0.8453 & 0.3321 \end{pmatrix}, \quad \cos\theta_6 = -0.1084.$$

It can be seen from the results that the second pair of solutions is the correct one. Normally, one can deduce the correct solution utilizing the local geodetic coordinates. Given the local geodetic coordinates as $\boldsymbol{r} = (\phi, \xi, r)^\top$, the geomagnetic potential is

$$V(r,\theta,\phi,q) = R_e \sum_{n=1}^{q} \left(\frac{R_e}{r}\right)^{n+1} \sum_{m=0}^{n} \left(g_n^m \cos m\phi + h_n^m \sin m\phi\right) P_n^m(\cos\xi) \tag{16}$$

where ϕ represents the longitude and ξ denotes the colatitude, which is the complement of the elevation angle (90° latitude). The term $P_n^m(\cos\xi)$ represents the Schmidt seminormalized associated Legendre functions of degree n and order m up to a maximum degree q, which form an orthogonal basis on the surface of a sphere. The coefficients g_n^m and h_n^m are the corresponding Gaussian coefficients that determine the contribution of each spherical harmonic term to the overall geomagnetic potential. The specific model used in this article is part of the international geomagnetic reference field (IGRF) series, which is typically released every five years to account for the secular variations in the Earth's magnetic field. Fig. 5 shows the geomagnetic total intensity all over the world at the sea level.

Using the geomagnetic potential in (16), one may obtain the vertical (Z) and horizontal (H) components of the Earth's magnetic field such that $Z = -(\partial V/\partial r)$ and $H = -(1/r)(\partial V/\partial \xi)$ with which the magnetic dip angle is obtained as

$$\theta = \arctan(Z/H). \tag{17}$$

One may acquire the approximate geodetic position of the calibration location and compute the dip angle or find out the approximate value in some lookup maps/tables.[1]

[1]https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml

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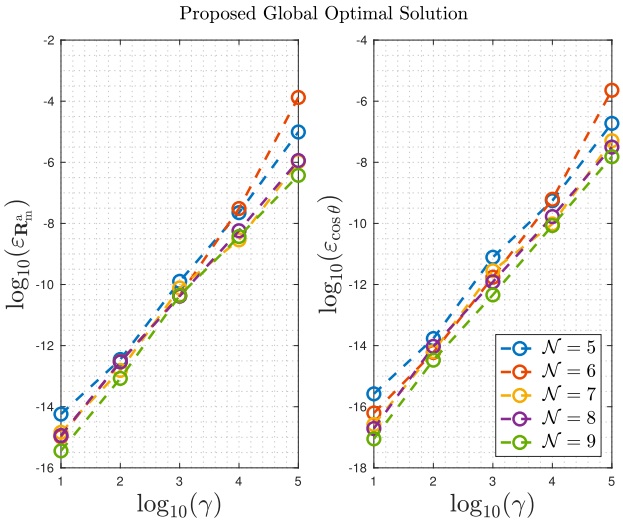


Fig. 6. Error statistics of AM-Align with increasing $\mathcal{N}$. All cases are convergent.

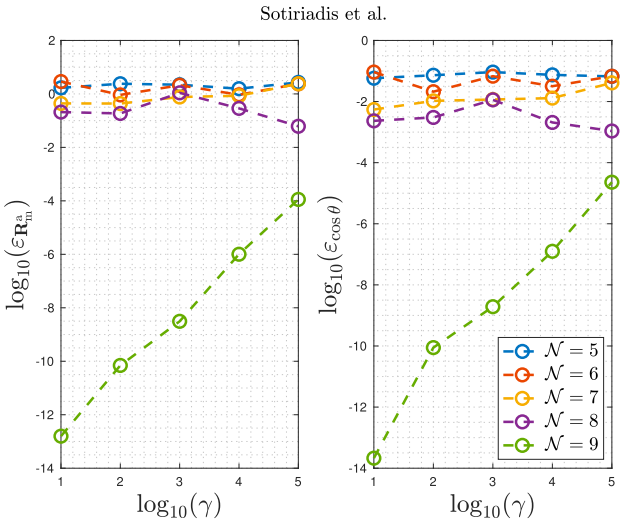


Fig. 7. Error statistics of the method by Sotiriadis et al. [14] with increasing measurement number $\mathcal{N}$. The results denote that the method by Sotiriadis et al. [14] achieves divergent estimates when $\mathcal{N} < 9$.

2) Monte Carlo Analysis: Multiple Monte Carlo simulations are performed with different numbers of measurements for AM-Align and the existing method by Sotiriadis and Papafotis [15]. Here, it should be noted that the work in [15] is a refined version using the result as an initial guess based on what they proposed previously in [14]. Therefore, the initial guess logics for the two works are consistent. We simulate over 10 000 samples for each case with different $\mathcal{N}$'s and noise levels γ and get mean results. The error metrics are

$$\varepsilon_{\boldsymbol{R}_m^a} = \arccos \frac{\mathrm{tr}\left(\boldsymbol{R}_m^a \boldsymbol{R}_{\mathrm{true}}^\top\right) - 1}{2}$$
$$\varepsilon_{\cos\theta} = |\cos\theta - \cos\theta_{\mathrm{true}}|.$$

The results are shown in Figs. 6 and 7, respectively, where $\log_{10}(\gamma)$ denotes noise levels of 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, and 10^{-1}. The results indicate that AM-Align requires less numbers of measurements. We also inspect the error results of various candidates when $\mathcal{N} = 9$, which are depicted in Fig. 8. The results also indicate that the proposed method, according to its globally optimal nature, has better estimation accuracy than the compared representative.

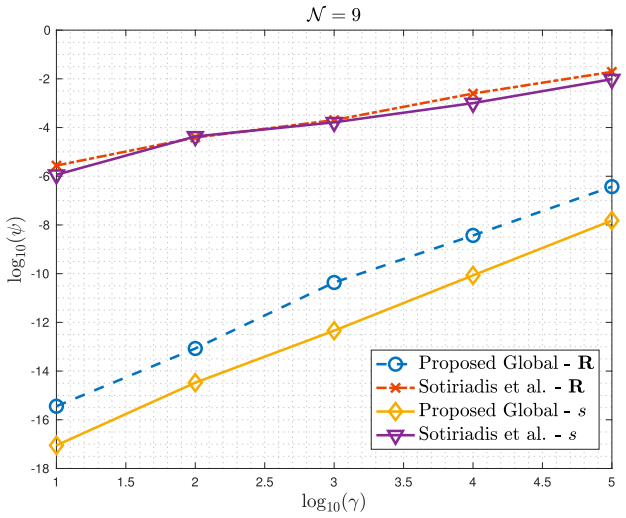


Fig. 8. Error statistics of various algorithms when $\mathcal{N} = 9$, in which ψ denotes the error metrics of either $\varepsilon_{\boldsymbol{R}_m^a}$ or $\varepsilon_{\cos\theta}$. Still, AM-Align has better estimation accuracy.

In fact, from Fig. 6, one may notice that the calibration accuracy would not have a very evident rise with more pairs of measurements. This is because, statistically, when $\mathcal{N}$ increases, the estimation error would be limited by the Cramér–Rao lower bound [27], which can easily be approximated with several statistical samples. That is to say, there is no need to gather too many measurement samples for an ultimate calibration precision. Instead, one needs to pay more attention to the data quality of the in-run sensor readings. Please refer to Section III-C for engineering tips.

For the uncertainty characterization of the proposed method, according to its high-nonlinear nature, it cannot be easily evaluated analytically. However, based on the error analysis shown in Fig. 6, one may come to a conclusion that for consumer-grade sensors with a sensor noise level around $10^{-2}\sim10^{-1}$, the rotation error would be around $10^{-8}\sim10^{-4}$ rad for $\mathcal{N} = 5$. It shall be noted that the estimation errors are quite small even with large sensor errors. Similar behavior happens for the method of Papafotis and Sotiriadis [14] when $\mathcal{N} \geq 9$. Thus, all compared methods are of high accuracy when they reach their respective minimum $\mathcal{N}$. The advantage of the proposed method would be better in the condition that $\mathcal{N} \leq 9$, which commonly happens in tasks with limited motion ability, e.g., heavy automotives and autonomous vehicles.

B. Real-World Experiments

1) Sensor Calibration: Before conducting the real-world experiments, we performed a thorough intrinsic calibration of the accelerometer and magnetometer using the Riemannian optimization approach proposed in [28] for the microaerial vehicle platform in Fig. 3. This calibration process estimated and compensated for the sensor-specific errors, such as scale factors, biases, and nonorthogonality, ensuring that the sensor

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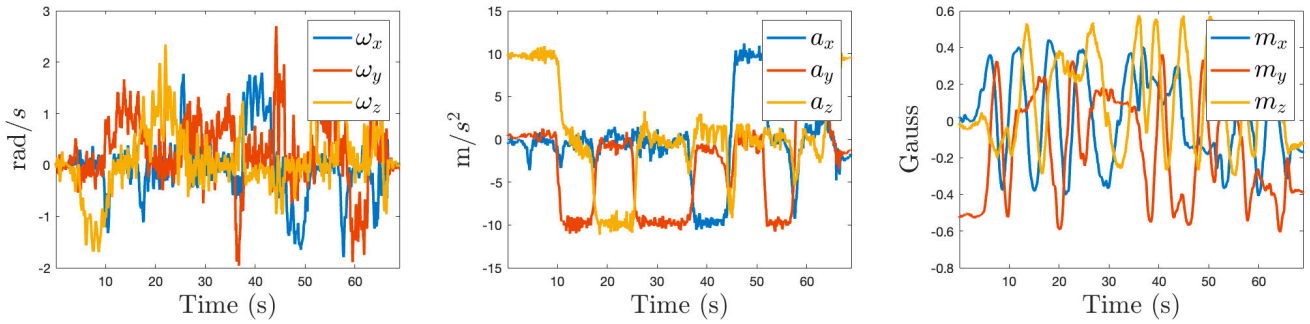


Fig. 9. Raw sensor outputs from the employed 3DM-GQ7 integrated navigation system. Angular velocity (left), acceleration (middle), and magnetic field (right).

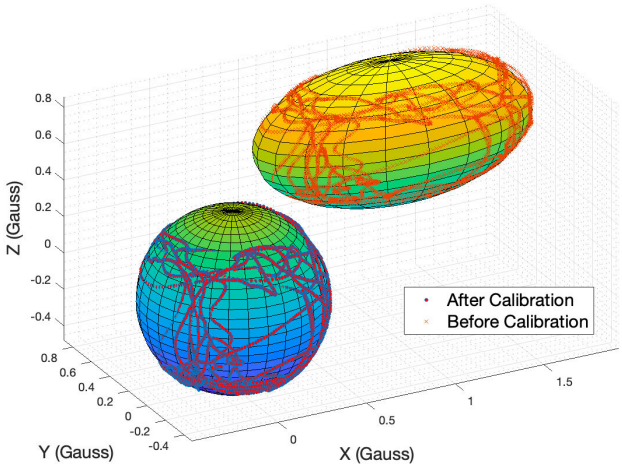


Fig. 10. Ellipsoid fitting to magnetometer observations before and after intrinsic calibration. Measurements well-distributed on a unit sphere, radius equal to magnetic field magnitude, after calibration (left). Uncalibrated data exhibit ellipsoidal distribution, demonstrating the efficacy of intrinsic calibration. The color gradient indicates increasing Z-axis values (Gauss) (right).

measurements were as accurate as possible. Fig. 10 illustrates the calibration results, showing the raw and calibrated magnetometer measurements. The calibrated measurements are well-distributed on a unit sphere, indicating the effectiveness of the calibration procedure in removing distortions and biases. This calibration step is crucial for the accurate evaluation of the AM-Align method, as it eliminates the influence of sensor-specific errors on the misalignment estimation process. In addition, in the experiments with the handheld platform illustrated in Fig. 3, both INS systems were precalibrated for magnetometer intrinsic parameters using the manufacturer software; hence, further discussion on this aspect is omitted.

2) Handheld Device Experiments: As presented in Fig. 3, a handheld platform with rigidly installed integrated navigation systems: the Lord 3DM-GQ7 and SBG Ellipse-N. Each system has an output rate of 500 Hz. The bias instabilities of the accelerometers are 5 and 14 μg for the 3DM-GQ7 and SBG Ellipse-N, respectively, while the magnetometer bias instability is 1.5°/h for the SBG Ellipse-N. As the two systems are synchronized, we verify the accuracy of the proposed AM-Align method by comparing the misalignment between the 3DM accelerometer and SBG magnetometer, with the gyro

TABLE I
ERROR EVALUATION OF $\varepsilon_{\boldsymbol{R}_g}$

	Case 1	Case 2
$\varepsilon_{\boldsymbol{R}_g}$	0.4692°	0.2833°

alignment $\boldsymbol{R}_g$ results between the two systems, such that

$$\underset{\boldsymbol{R}_g\in\mathrm{SO}(3),\boldsymbol{t}\in\mathbb{R}^3}{\arg\min}\sum\left\|\boldsymbol{\omega}_{\mathrm{3DM}}-\boldsymbol{R}_g\boldsymbol{\omega}_{\mathrm{SBG}}-\boldsymbol{t}\right\|^2 \tag{18}$$

in which $\boldsymbol{\omega}$ denotes angular velocity terms measured from two independent systems, while $\boldsymbol{t}$ specifies the extrinsic translation, which is not considered in this work. Equation (18) is a 3-D registration problem, which is solved by [29]. The reason for using the gyro–gyro alignment as a reference is that both systems utilize very high-end gyroscopes, leading to highly accurate misalignment estimation between them. The AMA alignment errors for this scenario are evaluated using the metric

$$\varepsilon_{\boldsymbol{R}_g}=\arccos\frac{\mathrm{tr}\left(\boldsymbol{R}_m^a\boldsymbol{R}_g^\top\right)-1}{2}.$$

Two independent experiments are carried out in a magnetic disturbance-free room shown in Fig. 3. Captured raw data readings are depicted in Fig. 9, where, for simplicity, only the data information from 3DM-GQ7 is plotted. The error metrics are summarized in Table I. The results indicate that the proposed AM-Align can achieve the calibration accuracy of misalignment at the degree level. Since both the accelerometer and magnetometer, as aiding references operated in motion within attitude estimation can only achieve the accuracy of several degrees [30], the calibration errors are satisfactory.

3) Microaerial Vehicle Experiments: AM-Align is implemented on a microaerial vehicle (MAV) platform where the accelerometer and magnetometer are synchronized using hardware clocks and impulses. The accelerometer and magnetometer onboard are within the package of the low-cost STMicroelectronics LSM303D. The sensor package is mounted as part of the Pixhawk autopilot 2.4.6, whose schematic is shown in https://github.com/pixhawk/Hardware/blob/master/FMUv2/PX4FMUv2.4.6.sch. The specifications of the sensor combination are listed in Table II. To improve the performance of the in-run data collection, following a common aerospace-engineering practice, we use the shock-absorbing sponge

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TABLE II
SPECIFICATIONS OF THE LOW-COST LSM303D SENSOR IN THE MAV TEST

Items	Values and Units
Acceleration Range	± 8g
Magnetometer Range	± 16 Gauss
Acceleration Sensitivity	0.244 mg/LSB
Magnetometer Sensitivity	0.479 mGauss/LSB
Acceleration Noise	150 ug/$\sqrt{\text{Hz}}$
Magnetometer Noise	5 mGauss/RMS

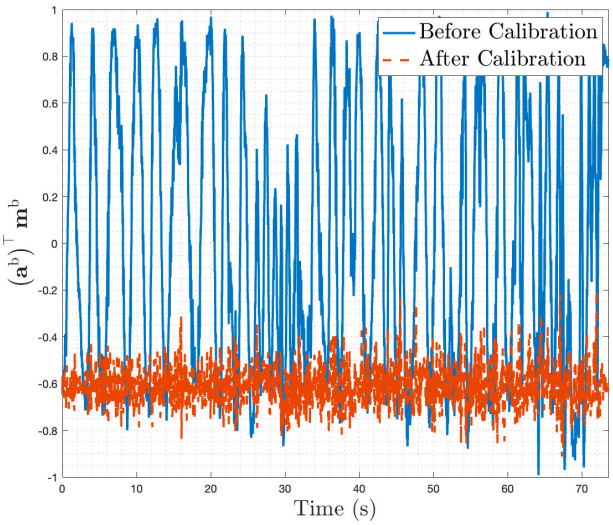


Fig. 11. $(\boldsymbol{a}^b)^\top \boldsymbol{m}^b$ values before and after the calibration using the proposed globally optimal method.

to reduce the in-flight high-frequency jitter. After intrinsic calibration in Fig. 11, AM-Align is used for misalignment calibration. In this case, no ground-truth information is available for evaluating the errors of the estimated misalignments. Therefore, indirectly, we compare the dot product $(\boldsymbol{a}^b)^\top \boldsymbol{m}^b$ before and after the calibration (see Fig. 11). The results indicate that AM-Align is effective with the in-field robot as it significantly increases the consistency of $(\boldsymbol{a}^b)^\top \boldsymbol{m}^b$, which is the kernel metric for evaluation of the calibration.

C. Discussion

In fact, the stability and feasibility of our proposed work directly depend on the data quality of the accelerometer and magnetometer. In the first category of experiments, our employed sensor suite contains the Lord 3DM-GQ7 and SBG Ellipse-N as sensor packages for the accelerometer and the magnetometer. According to the reference manual, these sensor packages are temperature-calibrated and water-proof, which means that there would be no disturbance effects in terms of temperature, humidity, wind, and ambient particulate concentration. Considering that in all these conditions, the acceleration and magnetic field measurement would not be influenced, the stability and feasibility of the data quality would be guaranteed. We also suggest that in any hardware design of a compact accelerometer/magnetometer combination, the temperature calibration together with water-proof ability shall be prepared beforehand. This is actually a common practice that has been widely studied previously [17]. In a later category of experiments, it is shown that the proposed method is capable of estimating the alignment parameters within low-cost sensors in LSM303D. For stability of sensor data streams, it is recommended that in environments with high-frequency jitter, certain vibration-rejection procedures are performed. This can be achieved easily by employing shock-absorbing sponge or vibration-harvesting metal springs.

Our method does not rely on long-term data accumulation for ultimate calibration precision. Instead, it relies on several ad hoc measurements captured independently. The superior performance of AM-Align can be attributed to several key factors. First, the globally optimal formulation ensures that the best solution is found regardless of initialization, enhancing robustness and accuracy compared to local optimization methods. Second, by exploiting the algebraic structure of sensor measurement models, the alignment problem is efficiently solved using the polynomial eigenvalue technique. This algebraic approach enables the computation of all global optima, contributing to the strong performance of AM-Align. Moreover, AM-Align's ability to handle minimal measurement sets makes it highly practical and efficient for real-world applications.

IV. CONCLUSION

This article presented a globally optimal method for estimating the misalignment between an accelerometer and a magnetometer. The proposed AM-Align formulates the alignment problem as a constrained least-squares optimization in unit quaternion space, which is reduced to a system of multivariate polynomial equations and solved efficiently using the polynomial eigenvalue technique. Extensive simulations and real-world experiments demonstrated that our algorithm outperforms state-of-the-art methods in terms of accuracy and robustness under various conditions. The main limitation of AM-Align is its computational complexity, which grows exponentially with the number of measurements. Future work may focus on extending the proposed framework to handle more complex sensor models, investigating more efficient polynomial solver techniques, and integrating our method with other state estimation and sensor fusion techniques for more accurate and robust navigation systems.

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